ΠΑΝΕΠΙΣΤΗΜΙΟ ΠΑΤΡΩΝ
UNIVERSITY OF PATRAS
Τμήμα Ηλεκτρολόγων Μηχανικών & Τεχνολογίας Υπολογιστών

Robust Control Lab

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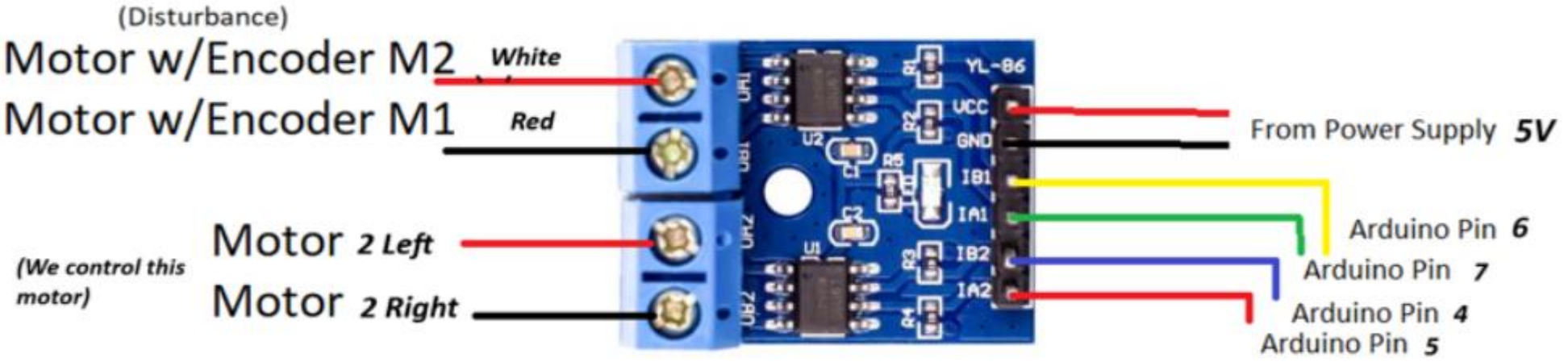
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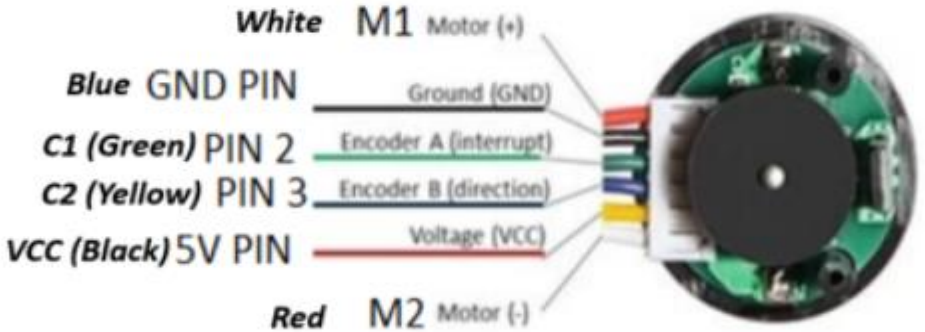
Velocity Control of a DC motor

Useful Information

- Password for Odroid: odroid
- Open Linux Terminal: CTRL+ALT+T
- Open Arduino IDE:
  - 1) Open new terminal
  - 2) cd ~/arduino-1.8.19/
  - 3) ./Arduino
- PWM to Voltage: V = PWM*(Vsupply)/255, Vsupply=5

Wiring


(Disturbance)
Motor w/Encoder M2
White
Motor w/Encoder M1
Red
(We control this motor)
Motor 2 Left
Motor 2 Right
YL-86
VCC
GND
IB1
IA1
IB2
IA2
From Power Supply 5V
Arduino Pin 6
Arduino Pin 7
Arduino Pin 4
Arduino Pin 5


White M1 Motor (+)
Blue GND PIN
Ground (GND)
C1 (Green) PIN 2
Encoder A (interrupt)
C2 (Yellow) PIN 3
Encoder B (direction)
VCC (Black) 5V PIN
Voltage (VCC)
Red M2 Motor (-)

ROS Information

- We will work with the robust_control_ws workspace.
- Inside robust_control_ws there is a single package (motor_controller) with a single node, (PID_Velocity_Controller)

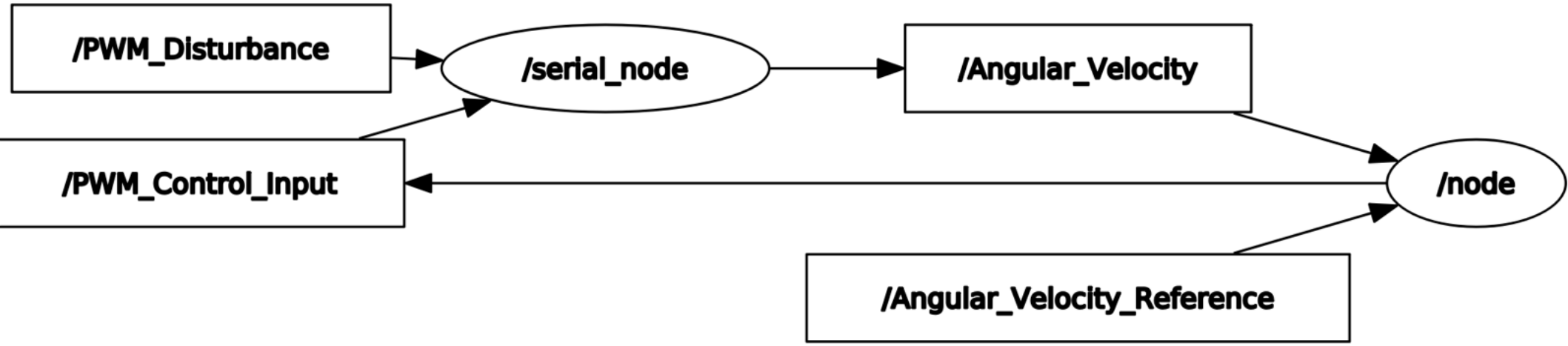
ROS Information

- Until now we ran:
  - roscore
  - rosrun rosserial_arduino serial_node.py /dev/ttyACM0
  - rosrun motor_controller PID_Velocity_Controller.py
  - rosrun rqt_plot rqt_plot
- You can run all the above with a single command:
  - roslaunch motor_controller Velocity_Control.launch

Note 1: The launch file is included in the launch directory of the motor_controller package

Note 2: The port in the launch file should match the port in the Arduino IDE (default is /dev/ttyACM0)

Velocity Control



Topics

- Angular_Velocity (RPM): Angular velocity value provided by the sensor
- Angular_Velocity_Reference (RPM): Reference signal for the angular velocity
- PWM_Control_Input : Control signal (Convert to voltage)
- PWM_Disturbance: Disturbance signal (Convert to voltage)
- Positive values in PWM indicate Positive angular velocity and CCW rotation. Negative values in PWM indicate negative angular velocity and CW rotation. (For the purposes of the exercise, the disturbance should be 'helping' the movement of the motor we aim to control)

Publish Data to a Topic

- Use rostopic:
- $ rostopic pub /topic_name std_msgs/type "data: value"
- Examples:
- rostopic pub /PWM_Disturbance std_msgs/Int16 "data: 100"
- rostopic pub /Angular_Velocity_Reference std_msgs/Float32 "data: 7.0" -r 10
- In rate mode, rostopic will publish your message at a specific rate. For example, -r 10 will publish at 10hz. Use rate mode to observe the desired state in the rqt_plot

‘Listen’ to Data that is being published

- Use rostopic:
- $ rostopic echo /topic_name

Example:

- rostopic echo /Angular_Velocity

Summary

Step 1 (Arduino): 1) CTRL+ALT+T 2) cd ~/arduino-1.8.19/ 3) ./arduino
4) Upload Robust_Control_DC_Motor.ino to Arduino (Choose correct port and model)

Step 2 (Launch File) 1) CTRL+ALT+T
2) roslaunch motor_controller Velocity_Control.launch

Step 3)(Publish Angular Velocity Reference Signal) 1) CTRL+ALT+T
2) rostopic pub /Angular_Velocity_Reference std_msgs/Float32 "data: 7.0" -r 10
(Change 7.0 to value of choice)

Step 4) (Publish Disturbance PWM) 1) CTRL+ALT+T 2)
rostopic pub /PWM_Disturbance std_msgs/Int16 "data: 100"
(Change 100 to value of choice, (Convert to voltage with V = PWM*(Vsupply)/255, Vsupply=5)

Step 5) Use rqt_plot (already included in the launch file) or
rostopic echo /Angular_Velocity to measure the angular velocity and
rostopic echo /PWM_Control_Input to measure the control input
(convert to voltage with V = PWM*(Vsupply)/255, Vsupply=5)